

2021? Mouser, for trailer

Model ENC-120-12



ENC Programmable Smart Charger Instruction Manual

2.1.1 LED indicators

LED	Description
Green(flashing)	Float (stage 3)
Orange(flashing)	Charging (stage 1 or stage 2)
None	Abnormal status (OTP or SCP)

2.2 Rear panel

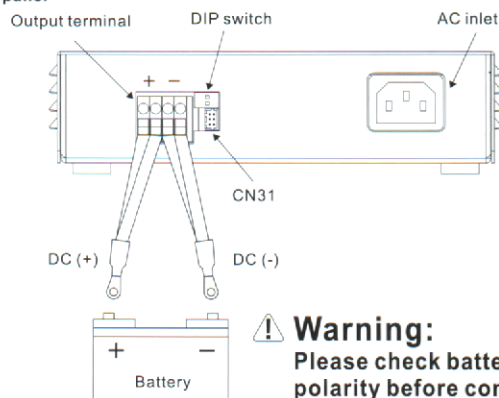


Figure 2.2 Rear panel of ENC charger

2.2.1 Pin assignment(CN31)

Pin No.	Function	Description
1	Prog- +3.3V	Installation for communication interface
2	Prog- GND	
3	Prog- RX	
4	Prog- TX	
5	RTH+	Installation for battery temperature sensor
6	RTH-	

2.2.2 Charging curve selection

ENC chargers provide a DIP switch on the rear panel used to switch different factory charging parameters. Please refer to the table below for corresponding curves on different positions of the DIP switch.

ON OFF

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0. Product Description

The ENC-120/180/360 family is Mean Well's new generation smart desktop charger, equipped with many of the protective features including battery over voltage, battery under voltage and reverse polarity. The latest high efficiency switching topology and microcontroller power management are utilized in their design. The family offers four selectable charging curves. With these pre-defined charging curves, the chargers are able to accommodate different battery brands and types, such as lead-acid batteries (Gel, flooded and AGM) or lithium-ion batteries (lithium-iron and lithium-manganese), by manually setting the DIP switch on the rear panel. In addition, users are allowed to build desired charging curves as well as adjusting parameters of the charger through the communication interface of the SBP-001 (not included in the package) to fulfill various specific charging applications. The factory charging curves are designed for lead-acid batteries. For other types of battery charging requirements, please contact Mean Well or our local distributors.

1. Safety Guidelines

- ⊙ Designed for charging lead-acid batteries (flooded, Gel and AGM) and some of the lithium chemistry batteries (lithium-iron and lithium-manganese).
- ⊙ The charger must be installed in a dry and well ventilated area. It should not be exposed to rain or open field.
- ⊙ The cables between charger and battery should be kept as short as possible to prevent excessive voltage drop (suggested cable length: 50cm - 100cm). Too much voltage drop will lead to longer charging period.
- ⊙ Temperature raising on case during charging is a normal behavior.
- ⊙ Dismantling or changing components of the charger is forbidden.
- ⊙ Make sure charging voltage and current meet battery specification.
- ⊙ Refrain from connecting new and old batteries in series.
- ⊙ Charger should be in the OFF mode before making battery connection or disconnection.
- ⊙ Three years warranty is provided under normal operating conditions. If failure results from improper operation, the warranty will not apply to the defective units.
- ⊙ This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions:
 - (a) This device may not cause harmful interference, and
 - (b) this device must accept any interference received, including interference that may cause undesired operation.

2. Appearance

2.1 Front panel

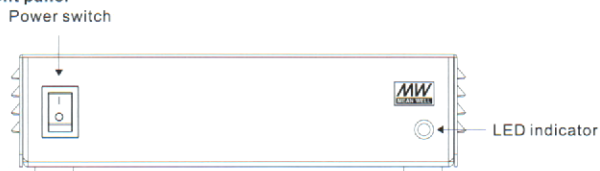


Figure 2.1 Front panel of ENC charger

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DIP SW position		12V model		
1	2	Description	Vboost	Vfloat
OFF	OFF	Default, programmable	14.4	13.8
ON	OFF	Pre-defined, gel battery	14.0	13.6
OFF	ON	Pre-defined, flooded battery	14.2	13.4
ON	ON	Pre-defined, AGM battery	14.5	13.5
DIP SW position		24V model		
1	2	Description	Vboost	Vfloat
OFF	OFF	Default, programmable	28.8	27.6
ON	OFF	Pre-defined, gel battery	28.0	27.2
OFF	ON	Pre-defined, flooded battery	28.4	26.8
ON	ON	Pre-defined, AGM battery	29.0	27.0
DIP SW position		48V model		
1	2	Description	Vboost	Vfloat
OFF	OFF	Default, programmable	57.6	55.2
ON	OFF	Pre-defined, gel battery	56.0	54.4
OFF	ON	Pre-defined, flooded battery	56.8	53.6
ON	ON	Pre-defined, AGM battery	58.0	54.0

NOTE: Voltage tolerance: $\pm 2\%$

3. Assembly Procedure

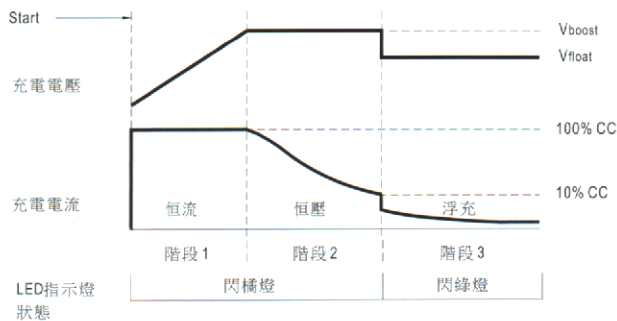
- ⊙ Choose a cable with suitable wire gauge according to the charging current to connect between the charger and the battery.
- ⊙ Please make sure the charger is OFF before connecting the battery to the output terminal. (Please note that there may be a small spark when connecting).
- ⊙ Battery polarity must be connected correctly: Terminal(+) to Battery(+); Terminal(-) to Battery(-), and take notice that the positive and negative polarities are not shorted.
- ⊙ Turn the charger ON after applying AC to the charger. If the LED indicator is flashing in orange, then the battery is being charged; if the LED is flashing in green, meaning the battery is already fully charged.

4. Operation Procedure

4.1 General operation

At the beginning stage of operation, the charger provides the largest current (e.g. 8 amps for the ENC-120-12) with 14.4Vdc of output voltage (e.g. default setting of the ENC-120-12) to charge batteries, the LED indicator will flash in orange. After a period of time (probably a couple of hours, based on the capacity of the batteries), the charging current will decrease gradually. After reducing to 10% of its maximum

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機型	恒流(Max.)	恒壓	浮充
ENC-120-12	8A	14.4V	13.8V
ENC-120-24	4A	28.8V	27.6V
ENC-120-48	2A	57.6V	55.2V
ENC-180-12	12A	14.4V	13.8V
ENC-180-24	6A	28.8V	27.6V
ENC-180-48	3A	57.6V	55.2V
ENC-240-12	16A	14.4V	13.8V
ENC-240-24	8A	28.8V	27.6V
ENC-240-48	4A	57.6V	55.2V
ENC-360-12	24A	14.4V	13.8V
ENC-360-24	12A	28.8V	27.6V
ENC-360-48	6A	57.6V	55.2V

圖4.1 三段式預設充電曲線設定值
NOTE: 電壓誤差: $\pm 2\%$, 電流誤差: 額定電流 $\pm 10\%$

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機型		恒壓		浮充		恒流		漸減電流	
		Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
ENC-120	12	9V	15V	9V	$\leq$ 恒壓	2.4A	8A	0.8A	2.4A
	24	18V	30V	18V	$\leq$ 恒壓	1.2A	4A	0.4A	1.2A
	48	36V	60V	36V	$\leq$ 恒壓	0.6A	2A	0.2A	0.6A
ENC-180	12	9V	15V	9V	$\leq$ 恒壓	3.6A	12A	1.2A	3.6A
	24	18V	30V	18V	$\leq$ 恒壓	1.8A	6A	0.6A	1.8A
	48	36V	60V	36V	$\leq$ 恒壓	0.9A	3A	0.3A	0.9A
ENC-240	12	9V	15V	9V	$\leq$ 恒壓	4.8A	16A	1.6A	4.8A
	24	18V	30V	18V	$\leq$ 恒壓	2.4A	8A	0.8A	2.4A
	48	36V	60V	36V	$\leq$ 恒壓	1.2A	4A	0.4A	1.2A
ENC-360	12	9V	15V	9V	$\leq$ 恒壓	7.2A	24A	2.4A	7.2A
	24	18V	30V	18V	$\leq$ 恒壓	3.6A	12A	1.2A	3.6A
	48	36V	60V	36V	$\leq$ 恒壓	1.8A	6A	0.6A	1.8A

NOTE: 電壓誤差: $\pm 2\%$, 電流誤差: 額定電流 $\pm 10\%$ (無論電流設定值是多少, 誤差都設定為額定電流 $\pm 10\%$), 例如ENC-120-12 額定電流8A, 誤差值則設定為0.8A, 若電流設定成3A, 則電流誤差範圍可能為3.8~2.2A。

6. 輸出連接線徑選擇

◎輸出連接線徑選擇: 電池充電時請依充電器充電電流大小選擇適當的線徑, 最小線徑需符合下表建議值:

AWG	CROSS SECTION(mm ²)	Max.Current(A) UL1015(600V 105°C)
12	3.309	22
14	2.081	12
16	1.309	8
18	0.823	6
20	0.517	4

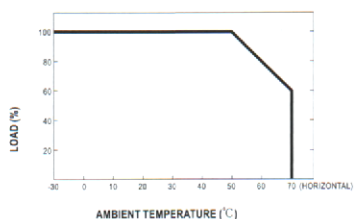
7. 建議電池容量選擇

充電器型號	建議電池容量
ENC-120-12	30-80Ah
ENC-120-24	15-40Ah
ENC-120-48	10-20Ah
ENC-180-12	45-125Ah
ENC-180-24	25-65Ah
ENC-180-48	13-35Ah
ENC-240-12	60-170Ah
ENC-240-24	30-85Ah
ENC-240-48	15-45Ah

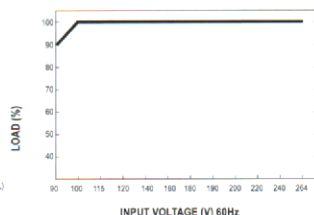
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4.2 減額曲線

4.2.1 充電電流與操作溫度



4.2.2 充電電流與輸入電壓



4.3 溫度補償

預設為-3mV/Cell/°C, 使用隨機附贈的NTC NSG05C250J5-500V連接至後面板通訊端子Rth+與Rth-。NTC感測線的線長可自行調整, 藉由連接需要的線長至NTC感測器部分與連結器部分。補償電壓如下表:

	Vboost		
	TA=0°C (RTH約17KΩ)	TA=25°C (RTH約5KΩ)	TA=50°C (RTH約1.73KΩ)
ENC-120-12	14.85V	14.4V	13.95V
ENC-120-24	29.7V	28.8V	27.9V
ENC-120-48	59.4V	57.6V	55.8V

NOTE電壓誤差: $\pm 2\%$

5. 自訂曲線及電池溫度補償

ENC可編程智能充電器提供使用者可程式化設定功能(需搭配MW_SBP-001智能充電器專用編程轉換器使用)。輸出電壓及充電電流範圍都可透過使用者需要自由進行調整與設定, 亦可調整電池溫度補償功能。
各機種電壓電流可調範圍如下表:

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充電器型號	建議電池容量
ENC-360-12	85-250Ah
ENC-360-24	45-125Ah
ENC-360-48	25-65Ah

備註:

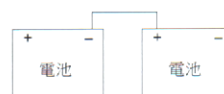
- 電池容量大於建議值並不會損壞電池, 只是電池充電時間需要長。
- 若您對電池可容許之充電電流有所疑問, 請參考電池廠商所提供之技術資料或詢問電池廠商。

8. 電池串聯及並聯

8.1 電池串聯

連接2顆電池串聯則電池提供電壓可增為2倍, 但AH容量維持不變。

例如: 將12V100AH規格的電池串聯2顆=24V100AH



8.2 電池並聯

連接2顆電池並聯則電池提供電壓維持不變, 但AH容量則變為2倍。

例如將12V100AH規格的電池並聯2顆=12V200AH



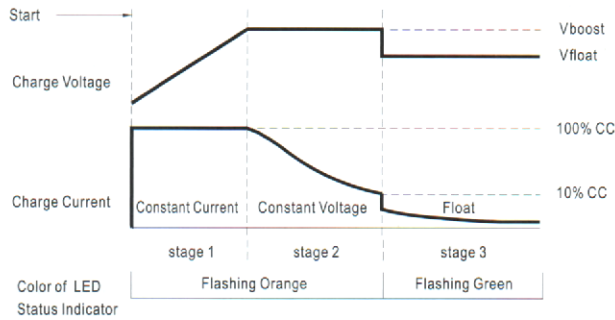
9. 故障排除

狀態	可能原因	排除方法
無法充電	ON/OFF 開關未切至ON	切換至ON位置
	輸入電壓過低	請確認輸入電壓應符合90~264VAC
長時間充電仍無法顯示綠燈	電池已老化或損壞	更換新電池
	輸出線材線徑太細	選擇適當線徑之線材

若仍無法排除故障情形, 請洽明緯或明緯經銷商

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value, the charger will go into "float charge" stage. Output voltage will drop to 13.8V (e.g. default setting of the ENC-120-12) and the LED indicator will flash in green. The relationship between charging current and charging voltage for each operation stage is shown in the curves below:



Model	Constant Current(Max.)	Constant Voltage	Float
ENC-120-12	8A	14.4V	13.8V
ENC-120-24	4A	28.8V	27.6V
ENC-120-48	2A	57.6V	55.2V
ENC-180-12	12A	14.4V	13.8V
ENC-180-24	6A	28.8V	27.6V
ENC-180-48	3A	57.6V	55.2V
ENC-240-12	16A	14.4V	13.8V
ENC-240-24	8A	28.8V	27.6V
ENC-240-48	4A	57.6V	55.2V
ENC-360-12	24A	14.4V	13.8V
ENC-360-24	12A	28.8V	27.6V
ENC-360-48	6A	57.6V	55.2V

Figure 4.1 Factory three stage charging curve setting

NOTE: Voltage tolerance: $\pm 2\%$, Current tolerance: Rated current $\pm 10\%$

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Model		Constant Voltage		Float Voltage		Constant Current		Tapper Current	
		Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
ENC-120	12	9V	15V	9V	$\leq CV$	2.4A	8A	0.8A	2.4A
	24	18V	30V	18V	$\leq CV$	1.2A	4A	0.4A	1.2A
	48	36V	60V	36V	$\leq CV$	0.6A	2A	0.2A	0.6A
ENC-180	12	9V	15V	9V	$\leq CV$	3.6A	12A	1.2A	3.6A
	24	18V	30V	18V	$\leq CV$	1.8A	6A	0.6A	1.8A
	48	36V	60V	36V	$\leq CV$	0.9A	3A	0.3A	0.9A
ENC-240	12	9V	15V	9V	$\leq CV$	4.8A	16A	1.6A	4.8A
	24	18V	30V	18V	$\leq CV$	2.4A	8A	0.8A	2.4A
	48	36V	60V	36V	$\leq CV$	1.2A	4A	0.4A	1.2A
ENC-360	12	9V	15V	9V	$\leq CV$	7.2A	24A	2.4A	7.2A
	24	18V	30V	18V	$\leq CV$	3.6A	12A	1.2A	3.6A
	48	36V	60V	36V	$\leq CV$	1.8A	6A	0.6A	1.8A

NOTE: Voltage tolerance: $\pm 2\%$, current tolerance: rated current $\pm 10\%$ (tolerance of the output current will be $\pm 10\%$ of the rated current no matter what current levels are set), e.g. rated current of ENC-120-12 is 8A, then there will be $\pm 0.8A$ tolerance on the setting of constant current and tapper current.

6. Output Wire Gauge Selection

⑤ Select an appropriate wire gauge based on rated charging current of the charger. The minimum requirements are in the table below.

AWG	CROSS SECTION(mm ²)	Max.Current(A) UL1015(600V 105°C)
12	3.309	22
14	2.081	12
16	1.309	8
18	0.823	6
20	0.517	4

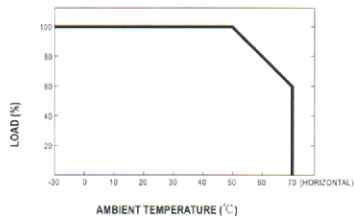
7. Recommended Capacity of Battery

Model	Suggested battery capacity
ENC-120-12	30-80Ah
ENC-120-24	15-40Ah
ENC-120-48	10-20Ah
ENC-180-12	45-125Ah
ENC-180-24	25-65Ah
ENC-180-48	13-35Ah
ENC-240-12	60-170Ah
ENC-240-24	30-85Ah
ENC-240-48	15-45Ah

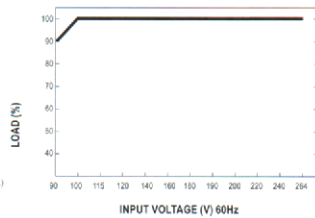
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4.2 Derating curve

4.2.1 Charging current VS ambient temperature



4.2.2 Charging current VS input voltage



4.3 Temperature compensation

The temperature sensor (NTC NSG05C250J5-500V) which comes with the charger can be connected to pin Rth+ and pin Rth- on CN31 of the rear panel to allow temperature compensation of the charging voltage. The wire length of the sensor can be adjusted according to different applications by linking the connector and sensor parts with wire length needed. Default setting is -3mV/Cell/°C, compensated voltages are shown as below:

	TA=0°C (RTH=17KΩ)	Vboost TA=25°C (RTH=5KΩ)	TA=50°C (RTH=1.73KΩ)
ENC-120-12	14.85V	14.4V	13.95V
ENC-120-24	29.7V	28.8V	27.9V
ENC-120-48	59.4V	57.6V	55.8V

NOTE: Voltage tolerance: $\pm 2\%$

5. Custom Charging Curve and Battery Temperature Compensation

The ENC chargers provide a programmable interface for the users (SBP-001, the Smart Battery Charging Programmer, is required for the function). All of the charging parameters, including constant current, constant voltage, float voltage and tapper current, and characteristics of battery temperature compensation can be adjusted and set.

Adjustable ranges for each model are shown as the table below:

Model	Suggested battery capacity
ENC-360-12	85-250Ah
ENC-360-24	45-125Ah
ENC-360-48	25-65Ah

Note:

- When the capacity of battery is larger than the recommended value, charging time will become longer but no harm will come to the battery.
- If there are doubts on the allowable charging current for the battery, please refer to the technical data provided by the battery manufacturer or inquire the battery manufacturer.

8. Series and Parallel Connection of Batteries

8.1 Batteries in series

Voltage can be doubled when 2 batteries are connected in series. However, the capacity (Ah) will remain the same.

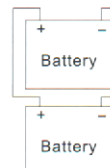
For example, 2 x 12V 100Ah batteries connected in series = 24V 100Ah.



8.2 Batteries in parallel

When 2 batteries are connected in parallel, voltage remains the same and the capacity (Ah) doubles.

For example, 2 x 12V 100Ah batteries connected in parallel = 12V 200Ah



9. Troubleshooting

Failure state	Possible Cause	Suggested Solutions
Unable to charge the battery	Power switch is in the OFF position	Switch to the ON position
	Input AC voltage is too low	Make sure input source is between 90~264VAC
LED indicator does not turn Green after a long charging period	Battery is old or damaged	Replace with a new battery
	Output cables are too thin	Replace with an appropriate wire gauge

If you are not able to clear the failure condition according to the instructions, please contact Mean Well or your nearest distributors for repair service.

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可編程智能充電器 使用手冊

2.2 後面板

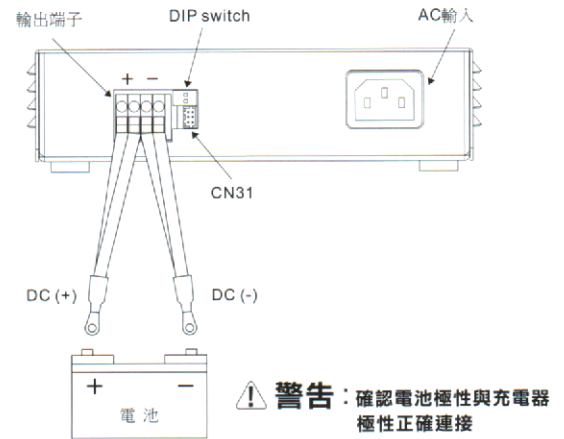


圖2.2 ENC充電器背面面板示意圖

2.2.1 連接器腳位與功能說明(CN31)



Pin腳編碼	功能	說明
1	Prog- +3.3V	通訊介面配件安裝
2	Prog- GND	
3	Prog- RX	
4	Prog- TX	
5	RTH+	電池溫度偵測配件安裝
6	RTH-	

2.2.2 曲線選取

ENC系列充電器的面板上提供一指撥開關(DIP SW)用以切換不同之預設充電參數，指撥開關(DIP SW)的開關切換位置與對應電壓如下：



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0. 產品簡述

ENC-120/180/240/360家族為明緯所推出新一代桌上型智慧型充電器，內建有電壓逆接、電池過壓、電池低壓保護等功能。設計上採用高效率硬體線路與微處理器電源管理，提供三段式(快充/均充/浮充)充電功能。充電器內建有四條預設充電曲線，使用者可透過充電器背面面板上的指撥開關調整或選擇適當的充電曲線進行充電，以因應不同廠牌或不同的電池種類(鉛酸電池：flooded加水式、Gel膠體式、AGM吸附玻璃纖維式等，或鋰電池族類：鋰鐵電池..等)所需求的特殊充電曲線調整。ENC充電器家族產品亦可透過本身內建的智慧型通訊介面，由使用者自行設定或修改充電參數與曲線，達到各種客製化之電池充電應用。您可針對某一區段的充電電壓、充電電流、截止電壓/電流等參數進行調整。本產品出廠設定均預設為鉛酸類蓄電池充電器，如有其他種類之充電應用請洽明緯。

1. 注意事項

- ◎適用於市面上常見鉛酸電池(flooded加水式、Gel膠體式、AGM吸附玻璃纖維式等)或部分鋰元素電池(鋰鐵電池)等。
- ◎充電時應置於室內通風良好，乾燥之場合使用，不可暴露於室外或雨中。
- ◎電池連接至充電器的導線長度儘可能短(建議約50cm~100cm)，避免因導線線路過大，拉長電池充電時間。
- ◎充電過程中，溫度升高為正常現象。
- ◎請勿拆解或變更產品上的任何零件。
- ◎使用前請先確認充電電壓和充電電流的設定符合電池要求。
- ◎電池串聯進行充電時，新舊電池不可混合做為串聯使用。
- ◎連接或拔除連接線時，請先確認充電器處於關機的狀態。
- ◎本充電器產品提供三年保固期，不當使用造成之損壞不在保固範圍內。
- ◎警告：ENC-120/180/240/360系列充電器符合BSMI認證，注意連接週邊設備需使用V1以上之防火外殼。

2. 面板說明

2.1 充電器正面

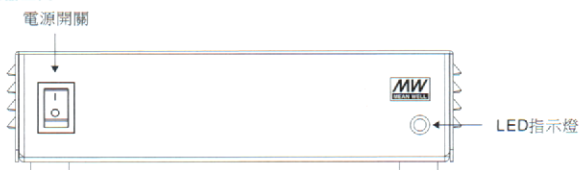


圖2.1 ENC充電器正面面板示意圖

2.1.1 LED指示燈號

LED燈色	狀態說明
綠燈(閃爍)	浮充段(stage 3)
橘燈(閃爍)	充電中(stage1 或 stage2)
無燈號	機器發生異常保護(過溫、短路)

①

DIPSW 位置		12V機型		
1	2	內建充電曲線說明	Vboost	Vfloat
OFF	OFF	Default, programmable	14.4	13.8
ON	OFF	Pre-defined, gel battery	14.0	13.6
OFF	ON	Pre-defined, flooded battery	14.2	13.4
ON	ON	Pre-defined, AGM battery	14.5	13.5
DIPSW 位置		24V機型		
1	2	內建充電曲線說明	Vboost	Vfloat
OFF	OFF	Default, programmable	28.8	27.6
ON	OFF	Pre-defined, gel battery	28.0	27.2
OFF	ON	Pre-defined, flooded battery	28.4	26.8
ON	ON	Pre-defined, AGM battery	29.0	27.0
DIPSW 位置		48V機型		
1	2	內建充電曲線說明	Vboost	Vfloat
OFF	OFF	Default, programmable	57.6	55.2
ON	OFF	Pre-defined, gel battery	56.0	54.4
OFF	ON	Pre-defined, flooded battery	56.8	53.6
ON	ON	Pre-defined, AGM battery	58.0	54.0

NOTE: 電壓誤差: $\pm 2\%$

3. 安裝步驟

- ◎依選用待充電之電池規格書所指示之充電電流選擇合適線徑之輸出線材，以作為連接電池與充電器使用。
- ◎確認充電器處於關機狀態下，將線材連接電池至輸出端子。(接上瞬間會有輕微火花產生屬於正常現象，請注意並放心使用)。
- ◎連接時電池與充電器的極性務必正確連接，端子台(+)接到電池(+)端，端子台(-)接到電池(-)端，並留意輸出正、負極間不可短路。
- ◎連接輸入線後，將ON/OFF開關切換至ON，此時LED指示燈會呈現橘燈閃爍(充電中)或綠燈閃爍(電池已充飽)。

4. 操作模式

4.1 一般操作狀態

於充電初期，充電器以最大電流(例:ENC-120-12預設為8A)對電池充電，充電電壓預設14.4V(例:ENC-120-12預設值)，當LED指示燈亮橘燈閃爍時，表示進入充電狀態中，充電經過一段時間後(時間長短需視選用之充電電池規格容量而定)，充電器輸出的充電電流會逐漸下降，當充電電流下降至最大充電電流規格值之10%(預設值左右時，充電器)會進入浮充階段，此時充電器輸出電壓將下降至浮充電壓值(例:ENC-120-12預設為13.8V)，此時LED指示燈亮綠燈並閃爍。各階段充電電流與充電電壓如下圖所示：

③